Spring 2021 Real Analysis II — Homework 5

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1. Let X and Y be Banach spaces and $T \in L(X, Y)$. Suppose T is surjective, and there exists $m > 0$ such that $\|Tx\| \geq m\|x\|$ for all $x \in X$. Show that T^{-1} exists, $T^{-1} \in L(Y, X)$, and $\|T^{-1}\| \leq 1/m$.

   Proof. Suppose $x, x' \in X$ and $Tx = Tx'$, then

   $$0 = \|Tx - Tx'\| = \|T(x - x')\| \geq m\|x - x'\|,$$

   which implies that $x = x'$. Hence T is also injective. Now we know T is bijective, and therefore T^{-1} exists and $T^{-1} \in L(Y, X)$. Moreover, for any $y \in Y$, we know there exists a unique $x \in X$, such that $Tx = y$, and thus

   $$\|T^{-1}y\| = \|x\| \leq \frac{1}{m}\|Tx\| = \frac{1}{m}\|y\|.$$

   Therefore $\|T^{-1}\| \leq 1/m$. □

2. Let X be a Hilbert space and $T \in L(X)$. Suppose there exists $m > 0$, such that $|\langle Tx, x\rangle| \geq m\|x\|^2$ for all $x \in X$. Show that T^{-1} exists and $T^{-1} \in L(X)$.

   Proof. It suffices to show that T is bijective. To see this, let $x, x' \in X$ such that $Tx = Tx'$, then

   $$0 = |\langle Tx - Tx', x - x'\rangle| = |\langle T(x - x'), x - x'\rangle| \geq m\|x - x'\|^2,$$

   which implies $x = x'$ and hence T is injective.

   Suppose $T(X) \neq X$, then there exists a nonzero $y \in T(X)^\perp$. Hence

   $$0 = |\langle Ty, y\rangle| \geq m\|y\|^2 > 0,$$

   which is a contradiction. Hence $T(X) = X$ and thus T is surjective. □

3. Let X and Y be B* spaces and D a linear subspace of X. Suppose $T : X \to Y$ is linear. Prove that the following statements hold:

   (a) If T is continuous and D is closed, then T is a closed operator.

   (b) Suppose T is continuous and closed. Then D is closed provided that Y is complete.

   (c) If T is closed and injective, then T^{-1} is closed.

   (d) If X is complete, T is closed and injective, $T(D)$ is dense in Y, and T^{-1} is continuous, then $T(D) = Y$.

   Proof. (a) Suppose that $\{x_k\} \subset D$ such that $x_k \to x$ and $Tx_k \to y$, then $x \in D$ since D is closed and $Tx = y$ since T is continuous. Hence by definition we know T is closed.

(b) Suppose $\{x_k\} \subset D$ and $x_k \to x$. Since T is continuous, we know $\{Tx_k\}$ is Cauchy. Since Y is complete, we know there exists $y \in Y$ such that $Tx_k \to y$. Hence $x \in D$ since T is closed.

(c) Suppose $\{y_k\} \subset D$, $y_k \to y$ and $T^{-1}y_k \to x$, then we know $x_k := T^{-1}y_k$ which satisfies $x_k \to x$ and $Tx_k = y_k \to y$. Since T is closed, we know $x \in D$ and $Tx = y$. This implies that $y \in D(T)$ and $x = T^{-1}y$. Therefore T^{-1} is closed.

(d) From (c) we know $T^{-1} : T(D) \to D$ is closed. Since T^{-1} is continuous and X is complete, from (b) we know that $T(D)$ is closed. Since $T(D)$ is dense in Y, we know $T(D) = Y$.

$\square$

4. Use Corollary 3.35 to show that $(C([0,1]), \|\cdot\|_1)$ is not a Banach space, where $\|f\|_1 := \int_0^1 |f(t)|\,\mathrm{d}t$.

Proof. Assume $(C([0,1]), \|\cdot\|_1)$ is a Banach space. Since $(C([0,1]), \|\cdot\|)$ is a Banach space, and

$$\|f\|_1 = \int_0^1 |f(t)|\,\mathrm{d}t \le \|f\|$$

where $\|f\| = \max_{0\le t\le 1} |f(t)|$. Then by Corollary 3.35, we know $\|\cdot\|_1$ and $\|\cdot\|$ are equivalent. However, define

$$f_k(t) = \begin{cases} k - k^2 t, & 0 \le t < 1/k, \\ 0, & 1/k \le t \le 1. \end{cases}$$

Then we can see $\|f_k\|_1 = 1$ but $\|f_k\| = k$ for all $k \in \mathbb{N}$, which is a contradiction. $\square$

5. Let $(X, \|\cdot\|)$ be a Banach space and $p : X \to \mathbb{R}$ satisfy

- (Nonnegative) $p(x) \ge 0$ for all $x \in X$.
- (Positive homogeneous) $p(\lambda x) = \lambda p(x)$ for all $\lambda > 0$ and $x \in X$.
- (Triangle inequality) $p(x_1 + x_2) \le p(x_1) + p(x_2)$ for all $x_1, x_2 \in X$.
- (Lower semicontinuous) If $x_k \to x$, then $\liminf_{k\to\infty} p(x_k) \ge p(x)$.

Show that there exists $M > 0$ such that $p(x) \le M\|x\|$ for all $x \in X$.

[Hint: define $\|x\|_1 := \|x\| + \max(p(x), p(-x))$, show that $(X, \|\cdot\|_1)$ is a Banach space, and then apply Corollary 3.35 to show that $\|\cdot\|_1$ and $\|\cdot\|$ are equivalent.

Proof. We first claim that $p(0) = 0$. To see this, we choose any nonzero $x_0 \in X$. Then $x_0/k \to 0$ as $k \to \infty$, and thus by lower semicontinuity of p we know

$$0 \le p(x) \le \liminf_{k\to\infty} p\left(\frac{x_0}{k}\right) = p(x_0) \lim_{k\to\infty} \frac{1}{k} = 0,$$

which implies $p(0) = 0$.

Now define $\|\cdot\|_1 := \|x\| + \max(p(x), p(-x))$. Then it is easy to show that $\|\cdot\|_1$ is a norm.

Then we shall show that $(X, \|\cdot\|_1)$ is complete. Let $\{x_k\}$ be a Cauchy sequence under $\|\cdot\|_1$. Then by the definition of $\|\cdot\|_1$ we know $\{x_k\}$ is Cauchy under $\|\cdot\|$. Since X is complete under $\|\cdot\|$, we know there exists $x \in X$ such that $\|x_k - x\| \to 0$. Moreover, for any $\epsilon > 0$, there exists $K = K(\epsilon) \in \mathbb{N}$, such that

$$\max(p(x_k - x_{k+j}), p(x_{k+j}) - p(x_k)) < \frac{\epsilon}{2}$$

for all $k \geq K$ and $j \in \mathbb{N}$. Therefore, we know by lower semicontinuity of p that

$$p(x_k - x) \leq \liminf_{j\to\infty} p(x_k - x_{k+j}) \leq \frac{\epsilon}{2}$$

for all $k \geq K$. Similarly, there is $p(x - x_k) \leq \epsilon/2$ for all $k \geq K$. Combining these, we know $\|x_k - x\|_1 \to 0$, which implies that $(X, \|\cdot\|_1)$ is a Banach space.

As $\|\cdot\|_1$ is stronger than $\|\cdot\|$, we know by Corollary 3.35 that they are equivalent. Hence there exists $M' > 0$ such that

$$\|x\|_1 = \|x\| + \max(p(x), p(-x)) \leq M'\|x\|,$$

which completes the proof by setting $M = M' - 1$. □

6. Let $l^1 := \{x = (\xi_1, \xi_2, \dots) : \sum_{k=1}^{\infty} |\xi_k| < \infty,\ \xi_k \in \mathbb{R}\}$ and $a = (a_1, a_2, \dots)$ a real sequence. Suppose $\sum_{k=1}^{\infty} a_k\xi_k$ converges for any $x \in l^1$. Show that $f : l^1 \to \mathbb{R}$ defined by $f(x) := \sum_{k=1}^{\infty} a_k\xi_k$ is a bounded linear functional, and $\|f\| = \sup_{k\in\mathbb{N}} |a_k|$.

Proof. Denote $\|x\|_1 := \sum_{k=1}^{\infty} |\xi_k|$ and $\|a\|_\infty = \sup_{k\in\mathbb{N}} |a_k|$. It is clear that f is a linear functional. Moreover

$$|f(x)| = \Big|\sum_{k=1}^{\infty} a_k\xi_k\Big| \leq \|a\|_\infty \sum_{k=1}^{\infty} |\xi_k| = \|a\|_\infty \|x\|_1.$$

Hence f is bounded and $\|f\| \leq \|a\|_\infty$.

For any $\epsilon > 0$, there exists $K = K(\epsilon) \in \mathbb{N}$, such that $\|a\|_\infty - \epsilon < |a_k| < \|a\|_\infty$. Let $x = \mathrm{sign} e_k$ where $e_k \in l^1$ is the unit vector with kth component being 1, then $f(x) = |a_k| > \|a\|_\infty - \epsilon$. Since $\epsilon > 0$ is arbitrary, we know $\|f\| \geq \|a\|_\infty$. □

7. Let X and Y be Banach spaces and D a linear subspace of X. Let $T : D \to T(D) \subset Y$ be a closed linear operator. Denote $Z := \{x \in D : Tx = 0\}$. Show that the following statements hold.

   (a) Z is a closed linear subspace of X.

   (b) $Z = \{0\}$ and $T(D)$ is closed in Y if and only if there exists $a > 0$ such that $\|x\| \leq a\|Tx\|$ for all $x \in D$.

Proof. (a) Let $\{x_k\}$ be a sequence in Z such that $x_k \to x$. Then, since $Tx_k = 0$ for all $k \in \mathbb{N}$ and T is closed, we know $Tx = 0$, i.e., $z \in Z$.

(b) ($\Leftarrow$) Since $Z = \{0\}$, we know $T : D \to T(D)$ is a one-to-one correspondence. Since $T(D)$ is closed and Y is complete, we know $T(D)$ is a Banach space. Let $\{y_k\}$ be a sequence in $T(D)$ such that $y_k \to y$, then $y \in T(D)$. Hence there exist x_k, x such that $Tx_k = y_k \to y = Tx$. On the other hand, suppose $x_k = T^{-1}y_k \to x'$. Then, since T is closed, we know $Tx' = y = Tx$, from which we deduce $x = x'$ since $Z = 0$. Therefore $T^{-1} : T(D) \to D$ is closed. By Closed Graph Theorem, $T^{-1} \in L(T(D), D)$. Hence there exists $a > 0$ such that for any $x \in D$, let $y = Tx$ and then there is

$$\|x\| = \|T^{-1}y\| \leq a\|y\| = a\|Tx\|.$$

($\Rightarrow$) If $Tx = 0$, then $\|x\| \leq a\|Tx\| = 0$ and thus $x = 0$. Hence $Z = \{0\}$. Let $\{x_k\}$ be a sequence in D such that $Tx_k \to y$ in Y. Hence $\{Tx_k\}$ is Cauchy. Thus

$$\|x_k - x_j\| \leq a\|Tx_k - Tx_j\| \to 0,$$

as $k, j \to \infty$. So $\{x_k\}$ is Cauchy in X. As X is a Banach space, there exists $x \in X$ such that $x_k \to x$ in X. Since T is closed, we know $x \in D$ and $Tx = y$, i.e., $y \in T(D)$. Hence $T(D)$ is closed.

$\square$

8. Let X and Y be Banach spaces and $T \in L(X, Y)$. Denote $Z := \{x \in X : Tx = 0\}$, $d(x, Z) := \inf_{z \in Z} \|x - z\|$, and

$$\gamma(T) := \inf_{x \notin Z} \frac{\|Tx\|}{d(x, Z)}.$$

Show that the following statements hold:

(a) $\gamma(T) \leq \|T\|$.

(b) If T^{-1} exists, then $\gamma(T) = \frac{1}{\|T^{-1}\|}$.

Proof. (a) For any $x \in X \setminus Z$ and $z \in Z$, we have $\|Tx\| = \|Tx - Tz\| \leq \|T\|\|x - z\|$. Taking infimum with respect to z on both sides yields $\|Tx\| \leq \|T\| d(x, Z)$. Note that $d(x, Z) > 0$ since Z is closed. Hence

$$\gamma(T) = \inf_{x \notin Z} \frac{\|Tx\|}{d(x, Z)} \leq \|T\|.$$

(b) If T^{-1} exists, then we know $Z = \{0\}$, $d(x, Z) = \|x\|$, and $\|T\| > 0$. Hence

$$\gamma(T) = \inf_{x \notin Z} \frac{\|Tx\|}{d(x, Z)} = \inf_{x \neq 0} \frac{\|Tx\|}{\|x\|} = \frac{1}{\sup_{x \neq 0} \frac{\|x\|}{\|Tx\|}} = \frac{1}{\sup_{y \neq 0} \frac{\|T^{-1}y\|}{\|y\|}} = \frac{1}{\|T^{-1}\|},$$

where we used change of variable $y = Tx$ since T is bijective.

$\square$

9. Let X be a Hilbert space and $a : X \times X \to \mathbb{R}$ a bilinear functional that satisfies

(a) There exists $M > 0$ such that $|a(x, y)| \leq M\|x\|\|y\|$ for any $x, y \in X$.

(b) There exists $\delta > 0$ such that $|a(x, x)| \geq \delta\|x\|^2$ for any $x \in X$.

Show that for any $f \in X^*$, there exists a unique $y_f \in X$ such that $a(x, y_f) = f(x)$ for any $x \in X$. Moreover y_f continuously depends on f.

Proof. By Lax-Milgram theorem, there exists $A \in L(X)$, where A is invertible and $\|A^{-1}\| \leq 1/\delta$, such that $a(x, y) = \langle x, Ay \rangle$ for all $x, y \in X$. For any $f \in X^* = X$ (we identify f with its counterpart in X using Riesz representation theorem), let $y_f = A^{-1}f$, then there is

$$a(x, y_f) = \langle x, Ay_f \rangle = \langle x, f \rangle = f(x), \quad \forall\, x \in X.$$

Since $A^{-1} \in L(X)$ we know $y_f = A^{-1}f$ depends on f continuously. $\square$

10. Let $\Omega \subset \mathbb{R}^2$ be an open bounded set with smooth boundary and $\alpha : \Omega \to \mathbb{R}$ a bounded measurable function satisfying $\alpha_0 \leq \alpha$ for some constant $\alpha_0 > 0$. Suppose $f \in L^2(\Omega)$ and denote

$$a(u, v) := \int_\Omega (\nabla u \cdot \nabla v + \alpha uv)\, \mathrm{d}x\, \mathrm{d}y \qquad \text{and} \qquad F(v) := \int_\Omega fv\, \mathrm{d}x\, \mathrm{d}y$$

for any $u, v \in H^1(\Omega)$. Show that there exists a unique $u \in H^1(\Omega)$ such that $a(u, v) = F(v)$ for any $v \in H^1(\Omega)$.

Proof. Let $X = H^1(\Omega)$. Since α is bounded, we know there exists $M > 0$ such that $|\alpha(x)| \le M$ for all $x \in \Omega$. Hence there are

$$a(u,v) \le \int_\Omega (|\nabla u||\nabla v| + M|u||v|)\,\mathrm{d}x \le \max(M,1)\|u\|_1\|v\|_1,$$
$$a(u,u) \ge \int_\Omega (|\nabla u|^2 + \alpha_0|u|^2)\,\mathrm{d}x \ge \min(\alpha_0,1)\|u\|_1^2.$$

Moreover, we know $F : X \to \mathbb{R}$ is linear and

$$F(v) = \int_\Omega fv\,\mathrm{d}x\,\mathrm{d}y \le \|f\|_{L^2}\|v\|_{L^2} \le \|f\|_{L^2}\|v\|_1,$$

and hence $F \in X^*$. By the problem above, we know there exists a unique u such that $a(u,v) = F(v)$ for all $v \in X$. $\square$